VPR Recording Guide

IE Tower — Computer Vision, Spring 2026

What are we doing

We're building an **indoor visual GPS** for the IE Tower. You take a photo anywhere in the building, the system tells you where it was taken (floor + specific area).

It works by comparing the photo against a database of images we record ourselves. Each image gets converted into a vector (a list of numbers) using a neural network. Similar places produce similar vectors. We use FAISS to search those vectors fast.

> The whole system is only as good as the data we collect. If the recordings are inconsistent or messy, the model won't work. That's why we all follow the same setup.

How the pipeline works

1. **We record video** of each location in the tower
2. **Split videos into clips** — one clip per location
3. **Extract frames** from each clip (1 frame per second)
4. **Generate embeddings** — ResNet50 turns each frame into a 2048-d vector
5. **Build a FAISS index** for fast similarity search
6. **Query** — upload a photo, get the closest matches + location

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Blackmagic Camera setup

Everyone uses **Blackmagic Camera** (free on the App Store, iOS). Download it before the recording session. Set everything manually — nothing on auto.

Setting	Value	Note
RESOLUTION	`720p`	Enough quality, small files
FRAME RATE	`30 FPS`	We extract 1fps anyway
CODEC	`HEVC`	H.265 — smaller files
LENS (OBJ)	`24mm`	Wide enough for indoors
SHUTTER (OBT)	`1/125`	Good balance for IE Tower lighting. Go 1/60 only in dark basements
ISO	`100`	Always 100 — adjust shutter instead
WHITE BALANCE	`~4300K`	Lock it 🔒 Indoor lighting
TINT	`0`	Neutral
FOCUS	`Manual`	Lock before each clip
HDR	`OFF`	No processing
ORIENTATION	`Horizontal`	Hold the phone sideways
CLIP LENGTH	`15-30 sec`	Per location

> The Blackmagic app might default to a higher white balance (like 5980K). That's too warm for IE Tower interiors. Bring it down to ~4300K and lock it.

> **Blown whites:** IE Tower has bright white walls + strong LED lighting. At f1.8 and 1/60 the highlights will clip easily (overexposed white areas with no detail). Fix this by raising the shutter speed — use 1/125 or 1/250 in bright areas like cafeterias, lobbies, and floors with big windows. Keep ISO at 100 always. Only raise ISO (up to 400 max) in dark basements or night recordings where even 1/60 isn't enough.

Recording tips

Before you press record

→ Verify all settings match the grid above

→ **Hold the phone sideways (landscape)** — physically rotate it so it's horizontal in your hands. Disabling vertical mode in the app makes the file landscape, but if you hold the phone upright the framing is sideways and useless. You need to actually hold it horizontal.

→ Lock focus on the main area before recording

→ Make sure the space looks normal — no unusual obstructions or events

While recording

→ **Walk slowly and steadily** — smooth movements, no shaking

→ **Cover the space** — pan gently so the model sees different angles

→ **Capture visual landmarks** — room numbers, signs, distinctive furniture, windows, structural details

→ **Avoid filming faces** — they add noise to the embeddings

→ **Vary the height** — record from different levels (standing, sitting, from stairs) so the model learns the same place from multiple perspectives. Just don't point straight at the ceiling or floor — keep the space visible.

→ **One clip = one location** — if you move to a new area, stop and start a new clip

Lighting conditions

Each location needs **at least 2 recordings** under different lighting:

→ **Day/afternoon** — natural light from windows, mixed lighting

→ **Night** — only artificial light, after dark or blinds closed

This teaches the model that the same place can look different depending on time of day.

Webcam footage (~10%)

For about 10% of the locations, also record a short clip with your **laptop webcam**. No special settings needed — that's the point. We want the model to handle lower-quality input from different cameras too. Just open Photo Booth or any recorder and film 10-15 seconds.

What happens after recording

Once you have your clips, the workflow is: split into individual clips → label each clip → extract frames → update the CSV → push to the repo. All the details for labeling and the folder structure are in the **Location Labels & Workflow** doc.